

0. Environment Setup & Automated Profiling

Before data cleaning, essential Python libraries were installed and verified to ensure a stable environment for dataset import, analysis, and automated profiling. This step ensured compatibility with Excel-based data files and enabled rapid exploration of dataset quality before manual cleaning.

```
pip install pandas ydata-profiling openpyxl xlrd
```

The following core packages were confirmed as installed: • pandas (v2.3.3) — for data manipulation and merging • ydata-profiling (v4.17.0) — for automatic data-quality and EDA reporting • openpyxl (v3.1.5) — for reading/writing Excel files • xlrd (v2.0.2) — for reading legacy .xls formats Supporting libraries such as numpy, matplotlib, scipy, and statsmodels were also verified.

Automated Dataset Profiling

After installation, automated HTML reports were generated for each dataset using ydata_profiling. These reports provided data type summaries, missing value detection, correlation matrices, and outlier visualizations, offering a comprehensive overview before deeper manual wrangling.

```
import os
import pandas as pd
from ydata_profiling import ProfileReport

# Set working directory
os.chdir("/Users/judycheng/Desktop")
print("■ Reports will be saved to:", os.getcwd())

# Supercharger dataset
supercharger_df = pd.read_excel("supercharger in washington state.xls")
ProfileReport(supercharger_df, title="■ Tesla Supercharger Network Profile (Washington)", explorativ

# Population dataset
population_df = pd.read_excel("Population 2024 age 25 to 59.xlsx")
ProfileReport(population_df, title="■ Washington Population Profile (Age 25-59)", explorativ

# EV VIN Registration dataset
ev_df = pd.read_excel("coordinates_output.xlsm")
ProfileReport(ev_df, title="■ EV VIN Registration Profile (Battery Electric Vehicles)", explorativ
```

Execution logs confirmed successful report creation for each dataset: ■ Reports will be saved to: /Users/judycheng/Desktop ■ Supercharger profile saved to Desktop ■ Population profile saved to Desktop ■ EV VIN registration profile saved to Desktop

1. Cleaning Steps Performed

- **File Import & Standardization:** Imported three datasets (Supercharger locations, County-level population by age group, and EV registration coordinates). Standardized column names and ensured consistent county/state naming formats.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import os
from openpyxl import load_workbook
from openpyxl.drawing.image import Image

super_df = pd.read_excel("/Users/judycheng/Desktop/supercharger in washington state.xls")
residents_df = pd.read_excel("/Users/judycheng/Desktop/Population 2024 age 25 to 59.xlsx")
ev_df = pd.read_excel("/Users/judycheng/Desktop/coordinates_output.xlsm")
```

- **Filtering by Region:** Filtered each dataset to include only rows where State == "Washington" and matched by county (King, Pierce, Kitsap, Chelan).
- **Data Type Conversion:** Converted numeric fields (population totals, area) to numeric formats and standardized text fields.
- **Merging Datasets:** Used county as the common key to compute per-county metrics such as EVs per charger and charger density.

2. Handling Missing Values

- **Supercharger Dataset:** Missing coordinates were rare; incomplete records were excluded.
- **Population Dataset:** Verified to have no missing entries.
- **EV Dataset:** Some missing county fields were filled using ZIP-based reverse geocoding or dropped if accuracy <95%.

```
ev_df = ev_df.dropna(subset=["County"])
```

3. Detecting & Handling Outliers

Used summary statistics (.describe()) and visualizations to spot extreme charger counts or EV totals. King County showed higher charger density, but this was expected from its urban population, so values were retained as valid variations rather than removed.

```
super_df.describe()
plt.hist(ev_df['some_numeric_column'], bins=20)
plt.show()
```

4. Subsetting Large Datasets

The EV registration dataset was large. To improve runtime, the analysis focused on county-level subsets (King, Pierce, Kitsap, Chelan), representing diverse urban, suburban, and rural areas.

```
subset_counties = ['King County', 'Pierce County', 'Kitsap County', 'Chelan County']
ev_subset = ev_df[ev_df['County'].isin(subset_counties)]
```

■ Summary

After all cleaning and profiling steps: • Datasets standardized by county and aligned by 2024 population, EV count, and charger count. • Missing values handled appropriately. • Outliers validated and retained as genuine regional variation. • The final dataset is ready for EV adoption modeling and charger build-out forecasting (2025–2050).